

Footprint Delta Trap [PakunFX]

```
// This Pine Script® code is subject to the terms of the Mozilla Public License 2.0 at https://mozilla.org/MPL/2.0/
// © PakunFX

//@version=6
indicator("Footprint Delta Trap [PakunFX]", shorttitle="FP TrapPRO", overlay=true,
max_labels_count=500, max_boxes_count=200, max_lines_count=500)

//=====
// Inputs
//=====
string groupFp = "Footprint"
int ticksPerRow = input.int(100, "Ticks per footprint row", minval=1, group=groupFp)
int valueAreaPct = input.int(70, "Value Area %", minval=1, maxval=100, group=groupFp)

string groupSignal = "Signal Logic"
int deltaLookback = input.int(20, "Delta average lookback", minval=5, group=groupSignal)
float deltaMult = input.float(1.5, "Strong delta multiplier", minval=0.1, step=0.1,
group=groupSignal)
float wickRatioMin = input.float(0.35, "Min wick ratio", minval=0.0, maxval=1.0, step=0.05,
group=groupSignal)
float bodyRatioMax = input.float(0.45, "Max body ratio", minval=0.05, maxval=1.0, step=0.05,
group=groupSignal)
bool showOnlyConfirmed = input.bool(true, "Show signals only on confirmed bars",
group=groupSignal)
bool showPreviewSignals = input.bool(true, "Show preview signals on forming bars",
group=groupSignal)

string groupLevels = "Levels / Zones"
bool showPOCZone = input.bool(true, "Show POC zone", group=groupLevels)
bool showVAZone = input.bool(true, "Show Value Area zone", group=groupLevels)
int zoneExtendBars = input.int(20, "Zone extend bars", minval=1, maxval=300, group=groupLevels)

string groupPanel = "Right Panel"
bool showPanel = input.bool(true, "Show right panel", group=groupPanel)
string panelTextSize = input.string("Small", "Panel text size", options=["Tiny", "Small", "Normal"],
```

```

group=groupPanel)
bool  showSignalRow  = input.bool(true, "Show signal row", group=groupPanel)
bool  showReasonRow  = input.bool(true, "Show reason row", group=groupPanel)
bool  showRatioRow   = input.bool(true, "Show candle ratio row", group=groupPanel)
bool  showPOCRow     = input.bool(true, "Show POC row", group=groupPanel)
bool  showVARow      = input.bool(true, "Show VA row", group=groupPanel)

string groupVisual = "Visual"
bool  showBg        = input.bool(true, "Background highlight on signal bars", group=groupVisual)
color bullColor     = input×color(color×lime, "Bullish color", group=groupVisual)
color bearColor     = input×color(color×red, "Bearish color", group=groupVisual)
color pocColor      = input×color(color×orange, "POC zone color", group=groupVisual)
color vaColor       = input×color(color×aqua, "VA zone color", group=groupVisual)
color panelHeaderColor = input×color(color×rgb(24, 119, 242), "Panel header color",
group=groupVisual)
color panelBgColor   = input×color(color×rgb(17, 24, 39), "Panel background color",
group=groupVisual)

//=====
// Helpers
//=====
f_getTextSize(_size) =>
    _size == "Tiny" ? size.tiny : _size == "Normal" ? size.normal : size.small

f_fmtPrice(_v) =>
    na(_v) ? "na" : str.toString(_v, format.mintick)

f_fmtVol(_v) =>
    na(_v) ? "na" : str.toString(_v, format.volume)

f_fmtPct(_v) =>
    na(_v) ? "na" : str.toString(_v, "#.###") + "%"

//=====
// Candle stats
//=====
float barRange      = math×max(high - low, syminfo.mintick)
float bodySize      = math×abs(close - open)

```

```
float upperWick    = high - math×max(open, close)
```

```
float lowerWick    = math×min(open, close) - low
```

```
float bodyRatio    = bodySize / barRange
```

```
float upperWickRatio = upperWick / barRange
```

```
float lowerWickRatio = lowerWick / barRange
```

```
//=====
```

```
// Footprint request
```

```
//=====
```

```
footprint fp = request×footprint(ticksPerRow, valueAreaPct)
```

```
float buyVol  = na
```

```
float sellVol = na
```

```
float deltaVol = na
```

```
float totalVol = na
```

```
float pocUp = na
```

```
float pocDn = na
```

```
float vahUp = na
```

```
float vahDn = na
```

```
float valUp = na
```

```
float valDn = na
```

```
bool fpReady = not na(fp)
```

```
if fpReady
```

```
    buyVol  := fp.buy_volume()
```

```
    sellVol := fp.sell_volume()
```

```
    deltaVol := fp.delta()
```

```
    totalVol := fp.total_volume()
```

```
    volume_row pocRow = fp×poc()
```

```
    volume_row vahRow = fp×vah()
```

```
    volume_row valRow = fp×val()
```

```
    pocUp := pocRow.up_price()
```

```
    pocDn := pocRow.down_price()
```

```
vahUp := vahRow.up_price()
vahDn := vahRow.down_price()
```

```
valUp := valRow.up_price()
valDn := valRow.down_price()
```

```
//=====
// Delta logic
```

```
//=====
float absDelta  = math.abs(deltaVol)
float avgAbsDelta = ta.sma(absDelta, deltaLookback)
```

```
bool strongPositiveDelta = fpReady and not na(avgAbsDelta) and deltaVol > avgAbsDelta *
deltaMult
```

```
bool strongNegativeDelta = fpReady and not na(avgAbsDelta) and deltaVol < -avgAbsDelta *
deltaMult
```

```
float deltaPct = fpReady and totalVol > 0 ? (deltaVol / totalVol) * 100.0 : na
```

```
//=====
// Trap logic
```

```
//=====
bool bearishTrapCore =
    strongPositiveDelta and
    upperWickRatio >= wickRatioMin and
    bodyRatio <= bodyRatioMax and
    close <= open
```

```
bool bullishTrapCore =
    strongNegativeDelta and
    lowerWickRatio >= wickRatioMin and
    bodyRatio <= bodyRatioMax and
    close >= open
```

```
bool priceRejectedAbovePOC =
    fpReady and not na(pocUp) and close < pocUp
```

```

bool priceHeldBelowPOC =
    fpReady and not na(pocDn) and close > pocDn

bool bearishTrap = bearishTrapCore and priceRejectedAbovePOC
bool bullishTrap = bullishTrapCore and priceHeldBelowPOC

//=====
// Confirmed / Preview signals
//=====
bool confirmedBullishTrap = barstate.isconfirmed and bullishTrap
bool confirmedBearishTrap = barstate.isconfirmed and bearishTrap

bool previewBullishTrap =
    showPreviewSignals and
    not barstate.isconfirmed and
    bullishTrap

bool previewBearishTrap =
    showPreviewSignals and
    not barstate.isconfirmed and
    bearishTrap

bool finalBullishTrap = showOnlyConfirmed ? confirmedBullishTrap : bullishTrap
bool finalBearishTrap = showOnlyConfirmed ? confirmedBearishTrap : bearishTrap

string signalText =
    finalBullishTrap ? "BUY" :
    finalBearishTrap ? "SELL" :
    previewBullishTrap ? "PRE-BUY" :
    previewBearishTrap ? "PRE-SELL" :
    "_"

color signalColor =
    finalBullishTrap ? bullColor :
    finalBearishTrap ? bearColor :
    previewBullishTrap ? color.new(bullColor, 0) :
    previewBearishTrap ? color.new(bearColor, 0) :
    color.silver

```

```

//=====
// Reason text
//=====
bool hasUpperWickCondition = upperWickRatio >= wickRatioMin
bool hasLowerWickCondition = lowerWickRatio >= wickRatioMin
bool hasBodyCondition      = bodyRatio <= bodyRatioMax

string reasonText = ""

if not fpReady
    reasonText := "No FP"
else if previewBullishTrap or finalBullishTrap
    reasonText := (strongNegativeDelta ? "-Δ " : "")
    reasonText += (hasLowerWickCondition ? "Wick " : "Need Wick ")
    reasonText += (priceHeldBelowPOC ? "POC" : "Need POC")
else if previewBearishTrap or finalBearishTrap
    reasonText := (strongPositiveDelta ? "+Δ " : "")
    reasonText += (hasUpperWickCondition ? "Wick " : "Need Wick ")
    reasonText += (priceRejectedAbovePOC ? "POC" : "Need POC")
else
    if strongPositiveDelta
        reasonText := "+Δ "
        reasonText += hasUpperWickCondition ? "" : "Need Wick "
        reasonText += hasBodyCondition ? "" : "Need Body "
        reasonText += priceRejectedAbovePOC ? "POC" : "Need POC"
    else if strongNegativeDelta
        reasonText := "-Δ "
        reasonText += hasLowerWickCondition ? "" : "Need Wick "
        reasonText += hasBodyCondition ? "" : "Need Body "
        reasonText += priceHeldBelowPOC ? "POC" : "Need POC"
    else
        reasonText := "Need Δ"

//=====
// Refined POC / VA styling
//=====
groupStyle = "POC / VA Style"

```

```

showRefinedLevels = input.bool(true, "Use refined POC / VA style", group=groupStyle)
showPOCGlow      = input.bool(true, "Show POC glow", group=groupStyle)
showVAFill       = input.bool(true, "Show VA fill", group=groupStyle)
levelExtendBars  = input.int(35, "Level extend bars", minval=5, maxval=300, group=groupStyle)
pocLineWidth     = input×int(2, "POC line width", minval=1, maxval=5, group=groupStyle)
vahValLineWidth  = input×int(1, "VAH/VAL line width", minval=1, maxval=3, group=groupStyle)

```

```

pocLineColor     = input×color(color×rgb(255, 70, 70), "POC line color", group=groupStyle)
pocGlowColor     = input×color(color×rgb(255, 70, 70), "POC glow color", group=groupStyle)
vaBorderColor    = input×color(color×rgb(180, 180, 180), "VA border color", group=groupStyle)
vaFillColor      = input.color(color.rgb(160, 160, 160), "VA fill color", group=groupStyle)

```

```
// POC center price
```

```

float pocMid = fpReady and not na(pocUp) and not na(pocDn) ? (pocUp + pocDn) * 0.5 : na
float vahMid = fpReady and not na(vahUp) and not na(vahDn) ? (vahUp + vahDn) * 0.5 : na
float valMid = fpReady and not na(valUp) and not na(valDn) ? (valUp + valDn) * 0.5 : na

```

```

var line pocLine = na
var line pocGlowTop = na
var line pocGlowBot = na
var line vahLine = na
var line valLine = na
var linefill vaFill = na
var box pocGlowBox = na

```

```
if barstate.islast and showRefinedLevels
```

```

    // cleanup old objects
    if not na(pocLine)
        line.delete(pocLine)
    if not na(pocGlowTop)
        line.delete(pocGlowTop)
    if not na(pocGlowBot)
        line.delete(pocGlowBot)
    if not na(vahLine)
        line.delete(vahLine)
    if not na(valLine)
        line.delete(valLine)
    if not na(vaFill)

```

```

    linefill.delete(vaFill)
if not na(pocGlowBox)
    box.delete(pocGlowBox)

// VA band
if fpReady and showVAZone and not na(vahUp) and not na(valDn)
    vahLine := line.new(
        x1 = bar_index,
        y1 = vahUp,
        x2 = bar_index + levelExtendBars,
        y2 = vahUp,
        color = color.new(vaBorderColor, 15),
        width = vahValLineWidth,
        style = line.style_solid)

    valLine := line.new(
        x1 = bar_index,
        y1 = valDn,
        x2 = bar_index + levelExtendBars,
        y2 = valDn,
        color = color.new(vaBorderColor, 15),
        width = vahValLineWidth,
        style = line.style_solid)

    if showVAFill
        vaFill := linefill.new(vahLine, valLine, color.new(vaFillColor, 82))

// POC center line
if fpReady and showPOCZone and not na(pocMid)
    pocLine := line.new(
        x1 = bar_index,
        y1 = pocMid,
        x2 = bar_index + levelExtendBars,
        y2 = pocMid,
        color = color.new(pocLineColor, 0),
        width = pocLineWidth,
        style = line.style_solid)

```



```

// soft POC glow as a narrow box
if showPOCGlow and not na(pocUp) and not na(pocDn)
    pocGlowBox := box.new(
        left = bar_index,
        top = pocUp,
        right = bar_index + levelExtendBars,
        bottom = pocDn,
        bgcolor = color.new(pocGlowColor, 78),
        border_color = color.new(pocGlowColor, 100))

//=====
// Signal plots
//=====
plotshape(
    finalBullishTrap,
    title = "Bullish Delta Trap",
    style = shape.triangleup,
    location = location.belowbar,
    color = bullColor,
    text = "BUY",
    textcolor = color×white,
    size = size×small)

plotshape(
    finalBearishTrap,
    title = "Bearish Delta Trap",
    style = shape.triangledown,
    location = location.abovebar,
    color = bearColor,
    text = "SELL",
    textcolor = color×white,
    size = size×small)

plotshape(
    previewBullishTrap,
    title = "Preview Bullish Delta Trap",
    style = shape.triangleup,
    location = location.belowbar,

```

```
color = color.new(bullColor, 65),
text = "PRE",
textcolor = color.white,
size = size.tiny)
```

```
plotshape(
    previewBearishTrap,
    title = "Preview Bearish Delta Trap",
    style = shape.triangledown,
    location = location.abovebar,
    color = color.new(bearColor, 65),
    text = "PRE",
    textcolor = color.white,
    size = size.tiny)
```

```
bgcolor(
    showBg ?
    finalBullishTrap ? color.new(bullColor, 88) :
    finalBearishTrap ? color.new(bearColor, 88) :
    previewBullishTrap ? color.new(bullColor, 93) :
    previewBearishTrap ? color.new(bearColor, 93) :
    na
    : na)
```

```
//=====
// Alerts
```

```
//=====
alertcondition(confirmedBullishTrap, title = "Bullish Delta Trap Confirmed", message = "PakunFX
Footprint Delta Trap PRO: Confirmed bullish trap detected")
alertcondition(confirmedBearishTrap, title = "Bearish Delta Trap Confirmed", message = "PakunFX
Footprint Delta Trap PRO: Confirmed bearish trap detected")
```

```
//=====
// Right panel table
//=====
var table fpPanel = table.new(position.top_right, 3, 10, border_width=1)
```

```
if barstate.islast and showPanel
```

```
table.clear(fpPanel, 0, 0, 2, 9)
```

```
txtSize = f_getTextSize(panelTextSize)
```

```
string fpStatusText = fpReady ? "● LIVE" : "● NO DATA"
```

```
color fpStatusBg = fpReady ? color.new(color.green, 25) : color.new(color.red, 25)
```

```
color deltaTextColor = na(deltaVol) ? color.white : deltaVol >= 0 ? bullColor : bearColor
```

```
color deltaPctColor = na(deltaPct) ? color.white : deltaPct >= 0 ? bullColor : bearColor
```

```
color signalBgColor =
```

```
    finalBullishTrap ? color.new(bullColor, 75) :
```

```
    finalBearishTrap ? color.new(bearColor, 75) :
```

```
    previewBullishTrap ? color.new(bullColor, 88) :
```

```
    previewBearishTrap ? color.new(bearColor, 88) :
```

```
    panelBgColor
```

```
    table.cell(fpPanel, 0, 0, "Item", text_color=color×white, text_size=txtSize,  
    bgcolor=panelHeaderColor)
```

```
    table.cell(fpPanel, 1, 0, "Value", text_color=color×white, text_size=txtSize,  
    bgcolor=panelHeaderColor)
```

```
    table.cell(fpPanel, 2, 0, "Reason", text_color=color×white, text_size=txtSize,  
    bgcolor=panelHeaderColor)
```

```
    table.cell(fpPanel, 0, 1, "FP Status", text_color=color×white, text_size=txtSize,  
    bgcolor=panelBgColor)
```

```
    table.cell(fpPanel, 1, 1, fpStatusText, text_color=color×white, text_size=txtSize,  
    bgcolor=fpStatusBg)
```

```
    table.cell(fpPanel, 2, 1, fpReady ? "Footprint OK" : "No footprint", text_color=color×white,  
    text_size=txtSize, bgcolor=panelBgColor)
```

```
    table.cell(fpPanel, 0, 2, "Buy Vol", text_color=color×white, text_size=txtSize,  
    bgcolor=panelBgColor)
```

```
    table.cell(fpPanel, 1, 2, f_fmtVol(buyVol), text_color=color×white, text_size=txtSize,  
    bgcolor=panelBgColor)
```

```
    table.cell(fpPanel, 2, 2, "Aggressive buy", text_color=color×white, text_size=txtSize,  
    bgcolor=panelBgColor)
```

```

    table.cell(fpPanel, 0, 3, "Sell Vol", text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)
    table.cell(fpPanel, 1, 3, f_fmtVol(sellVol), text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)
    table.cell(fpPanel, 2, 3, "Aggressive sell", text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)

    table.cell(fpPanel, 0, 4, "Delta", text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)
    table.cell(fpPanel, 1, 4, f_fmtVol(deltaVol), text_color=deltaTextColor, text_size=txtSize,
bgcolor=panelBgColor)
    table.cell(fpPanel, 2, 4, strongPositiveDelta ? "Strong +Δ" : strongNegativeDelta ? "Strong -Δ" :
"Weak Δ", text_color=deltaTextColor, text_size=txtSize, bgcolor=panelBgColor)

    table.cell(fpPanel, 0, 5, "Delta %", text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)
    table.cell(fpPanel, 1, 5, f_fmtPct(deltaPct), text_color=deltaPctColor, text_size=txtSize,
bgcolor=panelBgColor)
    table.cell(fpPanel, 2, 5, "Pressure ratio", text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)

    if showPOCRow
        table.cell(fpPanel, 0, 6, "POC", text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)
        table.cell(fpPanel, 1, 6, fpReady and not na(pocDn) and not na(pocUp) ? f_fmtPrice(pocDn) + "
- " + f_fmtPrice(pocUp) : "na", text_color=pocColor, text_size=txtSize, bgcolor=panelBgColor)
        table.cell(fpPanel, 2, 6, priceHeldBelowPOC ? "Bull support" : priceRejectedAbovePOC ? "Bear
reject" : "Neutral", text_color=color×white, text_size=txtSize, bgcolor=panelBgColor)
    else
        table.cell(fpPanel, 0, 6, "", bgcolor=panelBgColor)
        table.cell(fpPanel, 1, 6, "", bgcolor=panelBgColor)
        table.cell(fpPanel, 2, 6, "", bgcolor=panelBgColor)

    if showVARow
        table.cell(fpPanel, 0, 7, "VA", text_color=color×white, text_size=txtSize, bgcolor=panelBgColor)
        table.cell(fpPanel, 1, 7, fpReady and not na(valDn) and not na(vahUp) ? f_fmtPrice(valDn) + " -
" + f_fmtPrice(vahUp) : "na", text_color=vaColor, text_size=txtSize, bgcolor=panelBgColor)
        table.cell(fpPanel, 2, 7, "Value area", text_color=color×white, text_size=txtSize,

```

```

bgcolor=panelBgColor)
else
    table.cell(fpPanel, 0, 7, "", bgcolor=panelBgColor)
    table.cell(fpPanel, 1, 7, "", bgcolor=panelBgColor)
    table.cell(fpPanel, 2, 7, "", bgcolor=panelBgColor)

if showRatioRow
    string wickInfo = upperWickRatio >= wickRatioMin ? "Upper wick OK" : lowerWickRatio >=
wickRatioMin ? "Lower wick OK" : "Need wick"
    table.cell(fpPanel, 0, 8, "Body/Wick", text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)
    table.cell(fpPanel, 1, 8, "B:" + str.toString(bodyRatio, "#.##") + " U:" +
str.toString(upperWickRatio, "#.##") + " L:" + str.toString(lowerWickRatio, "#.##"),
text_color=color×white, text_size=txtSize, bgcolor=panelBgColor)
    table.cell(fpPanel, 2, 8, wickInfo, text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)
else
    table.cell(fpPanel, 0, 8, "", bgcolor=panelBgColor)
    table.cell(fpPanel, 1, 8, "", bgcolor=panelBgColor)
    table.cell(fpPanel, 2, 8, "", bgcolor=panelBgColor)

if showSignalRow
    table.cell(fpPanel, 0, 9, "Signal", text_color=color×white, text_size=txtSize,
bgcolor=panelBgColor)
    table.cell(fpPanel, 1, 9, signalText, text_color=signalColor, text_size=txtSize,
bgcolor=signalBgColor)
    table.cell(fpPanel, 2, 9, showReasonRow ? reasonText : "", text_color=color.white,
text_size=txtSize, bgcolor=panelBgColor)
else
    table.cell(fpPanel, 0, 9, "", bgcolor=panelBgColor)
    table.cell(fpPanel, 1, 9, "", bgcolor=panelBgColor)
    table.cell(fpPanel, 2, 9, "", bgcolor=panelBgColor)

if barstate.islast and not showPanel
    table.clear(fpPanel, 0, 0, 2, 9)

//=====
// Data Window

```

```
//=====
plot(fpReady ? deltaVol : na, "Footprint Delta", display=display.data_window)
plot(fpReady ? buyVol : na, "Footprint Buy Volume", display=display.data_window)
plot(fpReady ? sellVol : na, "Footprint Sell Volume", display=display.data_window)
plot(fpReady ? totalVol : na, "Footprint Total Volume", display=display.data_window)
plot(fpReady ? pocUp : na, "POC Upper", display=display.data_window)
plot(fpReady ? pocDn : na, "POC Lower", display=display.data_window)
plot(fpReady ? vahUp : na, "VAH Upper", display=display.data_window)
plot(fpReady ? valDn : na, "VAL Lower", display=display.data_window)
```